

Evaluation of the Impact of Indonesia's Government Regulation No. 36 of 2023 on Foreign Exchange Earnings from Natural Resource Exports (DHE SDA) on Changes in the Export Index from 2022 to 2023

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Abstract—This study investigates the impact of Indonesia's Government Regulation No. 36 of 2023 on Foreign Exchange Earnings from Natural Resource Exports (DHE SDA) by evaluating changes in the national export index between 2022, the pre-policy period, and 2023, the first year of implementation. The regulation aims to strengthen foreign exchange liquidity and support macroeconomic stability by requiring a greater portion of export proceeds to be held within the domestic financial system. To assess whether the policy corresponds with measurable shifts in export performance, this research employs the Mann-Whitney U test, a nonparametric approach suitable for export indicators that exhibit non-normal distributions and potential heteroskedasticity across years. The analysis focuses on detecting distributional differences rather than simple mean comparisons, providing a more robust understanding of policy effects under uncertain statistical conditions. The results indicate whether export index values in 2023 diverge significantly from those in 2022 and evaluate the reliability of this evidence through significance levels and effect-size measures. Overall, the study contributes to the empirical literature on trade policy and foreign exchange management by offering early insights into the effectiveness of PP 36/2023 in shaping short-term export outcomes and informing future regulatory refinement.

I. INTRODUCTION

The management of foreign exchange earnings generated from natural resource exports plays a central role in shaping the macroeconomic stability of emerging economies. Indonesia, whose export structure is heavily influenced by commodity based sectors, has long faced challenges associated with external vulnerability, including fluctuations in global commodity prices, episodes of exchange rate depreciation, and persistent pressures on the current account [2]. These pressures highlight the importance of ensuring that foreign exchange inflows remain available within the domestic financial system in order to support liquidity, strengthen financial intermediation, and safeguard macroeconomic stability. Within this context, the

Indonesian government introduced Government Regulation Number 36 of 2023 concerning Foreign Exchange Earnings from Natural Resource Exports, commonly referred to as PP 36 of 2023. The regulation came into effect on 1 August 2023 and represents a significant policy shift aimed at enhancing the management of export related foreign exchange resources [4].

PP 36 of 2023 contains two principal provisions that directly influence exporters in natural resource based industries, which include mining, plantations, forestry, and fisheries. First, exporters are required to channel their export proceeds into the Indonesian Financial System, known as Sistem Keuangan Indonesia or SKI. Second, for export values of at least USD 250000, exporters must allocate a minimum of 30 percent of their foreign exchange earnings into domestic financial instruments for a mandatory holding period of no less than three months. These provisions are designed to ensure that a significant share of foreign exchange inflows is retained within the country, thereby expanding domestic liquidity and supporting the stability of domestic money markets (Ministry of Finance, 2023). By increasing the availability of foreign exchange within the banking system, the regulation is also expected to contribute indirectly to the stability of the Indonesian rupiah, which remains sensitive to global economic conditions and capital flow dynamics.

Despite the strategic importance of PP 36 of 2023, limited empirical evidence exists regarding its short term impact on export performance. Most academic studies on trade and export policies concentrate on tariff reforms, non tariff measures, global competitiveness, or international trade agreements [6], [7]. Policies that specifically regulate the retention and placement of foreign exchange earnings have received comparatively little analytical attention, particularly in economies where natural resource exports form a large portion of total

foreign exchange inflows. Moreover, export indicators such as the export index tend to exhibit volatility, non linear patterns, and non normal statistical distributions due to external shocks, seasonal variations, and commodity market cycles [8]. These characteristics complicate the measurement of policy effects and necessitate analytical methods that do not rely on strong distributional assumptions.

Given these considerations, evaluating the impact of PP 36 of 2023 requires an approach that focuses on distributional differences in export outcomes rather than solely on mean changes. This study addresses this methodological need by applying the Mann Whitney U test, which provides a robust nonparametric comparison of export index values before and after the implementation of the regulation. The pre policy period represented by the year 2022 serves as the baseline, while the post policy period represented by the year 2023 reflects the initial phase of regulatory enforcement. Through this analytical framework, the study seeks to determine whether the distribution of Indonesia's export index in 2023 shows statistically significant divergence from that of 2022 and whether such divergence may be associated with the introduction of PP 36 of 2023.

By investigating the early economic effects of this regulation, the study contributes to the broader literature on foreign exchange management, export performance, and macroeconomic policy in resource dependent economies. The empirical insights derived from this research are expected to support policymakers in assessing the effectiveness of foreign exchange retention requirements and in refining regulatory mechanisms aimed at strengthening Indonesia's external resilience and financial stability.

II. LITERATURE REVIEW

The management of export proceeds and repatriation requirements has been a recurring policy tool in resource-dependent economies seeking to stabilize foreign exchange liquidity and the exchange rate. Empirical and policy work indicates that mandates or incentives to retain export earnings onshore can, at least temporarily, increase the supply of foreign currency available to domestic markets and thereby contribute to short-run exchange rate and liquidity objectives [16]. However, the magnitude and persistence of these effects vary across countries and institutional settings, and studies of capital-account measures that include restrictions on repatriation show heterogeneous effects on capital flows and financial variables, implying that policy design and the wider macrofinancial environment matter for outcomes [17].

Indonesia's Government Regulation No. 36 of 2023 (PP 36/2023) formalizes requirements for exporters in natural resource sectors (mining, plantations, forestry, fisheries) to channel export proceeds into the Indonesian financial system and, for documents with export value $\geq$ USD 250,000, to place at least 30 percent of proceeds in domestic instruments for a minimum of three months [4]. The regulation became effective in 2023 and has been complemented by Bank Indonesia

implementing rules (PBI) that define placement instruments, reporting procedures, and compliance mechanisms [12]. Initial government statements emphasize that the policy aims to strengthen domestic foreign exchange liquidity and indirectly support rupiah stability [13].

While the policy objective of improving domestic FX liquidity is clear, early reactions from exporters and trade associations highlight concerns about cash-flow impacts and operational burdens, especially when retention requirements are tightened [15]. Press coverage and government updates through 2024 and 2025 also document ongoing implementation adjustments and evaluations of compliance modalities [14]. These observations underscore that real-world effects depend on enforcement, the availability of attractive domestic FX instruments, and firms' financing structures, which means that repatriation rules may have unintended consequences if domestic markets cannot efficiently absorb retained FX.

Recent empirical work specifically examining export proceeds repatriation in Indonesia and comparable settings provides nuanced findings. A 2025 study employing time-varying parameter models finds that repatriated export proceeds have horizon-dependent effects on exchange rate volatility, with limited buffering power against short-term capital flow shocks in many periods and more meaningful effects during earlier phases of enforcement [16]. The authors also emphasize the importance of complementary market development, including deeper FX instruments and hedging markets, for sustained policy effectiveness. This literature suggests that single-period before-after comparisons may reflect short-run distributional changes in export outcomes but cannot, on their own, establish persistent causal effects without accounting for broader confounders.

Methodologically, evaluating pre-post differences in export indicators requires attention to data distribution and sample structure. Export indices and export-value time series frequently show volatility, skewness, and seasonality; when normality and homoscedasticity assumptions are questionable, nonparametric rank-based tests such as the Mann-Whitney U test for independent samples or the Wilcoxon signed-rank test for paired observations provide robust alternatives to parametric t-tests [10], [11]. Standard texts on nonparametric inference recommend supplementing hypothesis tests with effect-size measures and bootstrap confidence intervals to evaluate the magnitude and robustness of detected differences. Given the short post-policy horizon for PP 36/2023, a non-parametric framework with robustness checks (such as controls for commodity-price shocks, exchange-rate conditions, and the staggered timing of PBI implementation) is suitable for an initial assessment of distributional shifts in the export index between 2022 and 2023.

Taken together, the reviewed policy literature and empirical findings imply three lessons relevant to the present paper. First, repatriation or placement requirements can influence domestic FX liquidity, but effects are context-dependent and often temporary [16]. Second, implementation details and firm-level financial structures strongly mediate observed out-

comes [14]. Third, rigorous empirical design is necessary: while a nonparametric pre–post comparison can detect short-term distributional changes, it should be complemented by robustness checks and structural context to avoid attributing all changes solely to the regulation.

III. METHODOLOGY

A. Data Collection and Description

The data used in this study were obtained from the official publications of Statistics Indonesia (Badan Pusat Statistik, BPS), specifically the annual Export Index of Raw Materials and Manufactured Products for 2022 and 2023. These datasets include sector codes, sector descriptions, and export index values that reflect annual export performance. Each file was examined to understand its variable structure, consistency across years, and any documented revisions in sector classifications.

The datasets were downloaded in spreadsheet format and archived to ensure reproducibility. A preliminary assessment was carried out to verify that sector definitions and index calculation methods remained comparable between 2022 and 2023. Any inconsistencies or data gaps were documented prior to cleaning.

B. Data Cleaning and Preparation

Before analysis, the raw datasets underwent a comprehensive cleaning process. The data were imported into statistical software and checked for duplicate entries, incorrect formats, missing values, and inconsistencies across variables. Duplicated observations with identical sector codes, years, and index values were removed. Missing values were reviewed sector by sector, and sectors with substantial missingness were flagged.

All numerical variables were standardized into a consistent format. The data were then reorganized into a panel structure with the variables sector code, year, and export index. A clean version of the dataset was saved for the next analytical steps. This preparation step ensures that index values for both years are aligned for comparison.

C. Sector Filtering

The analysis focuses on sectors coded 0, 1, 2, 3, 4, 6, 8, and 9. These sectors represent key export categories relevant to the purpose of the study. From the cleaned dataset, only observations belonging to these selected sectors were retained. A verification step ensured that each sector contained data for both 2022 and 2023.

This filtering produced a compact sectoral panel that is suitable for descriptive and inferential analysis. All excluded data and the reasons for exclusion were documented to ensure transparency.

D. Descriptive Statistics

Descriptive statistics were calculated to provide an initial picture of export index performance. For every selected sector, the mean and median values were computed for 2022 and 2023. These values illustrate how export indices shifted before and after the implementation of PP 36 of 2023.

A summary table was developed to compare the means and medians across years and to calculate the differences between them. These descriptive results reveal whether sectoral export indices generally increased or decreased following the policy introduction. Visual tools such as boxplots may also be applied to identify skewness, distribution shape, and potential outliers. This step helps justify the choice of nonparametric tests in the inferential stage.

E. Nonparametric Statical Testing

Given the limited number of observations and the possibility that distributional assumptions may not hold, the Mann Whitney U test was selected to examine whether the 2023 export index values differ significantly from those of 2022.

- 1) Import the cleaned dataset.
- 2) Navigate to Analyze, Nonparametric Tests, Legacy Dialogs, and 2 Independent Samples.
- 3) Select export index as the test variable and year as the grouping variable.
- 4) Run the test separately for each sector by applying a filter or grouping structure.

For each sector, the U statistic, p value, and sample size were recorded. Although the short time span limits statistical power, this nonparametric test provides a robust framework for preliminary distributional comparison between the two years.

F. Comparative Interpretation of Descriptive and Inferential Results

After completing both descriptive and inferential analyses, the results were compared to evaluate consistency. A summary table was created showing mean and median differences between 2022 and 2023 alongside the Mann Whitney U test outcomes. This comparison helps determine whether statistically significant differences support or contradict the descriptive findings.

Sectors that showed statistically significant changes were further examined in terms of direction and magnitude. For sectors without significant differences, the findings were interpreted cautiously, considering external influences such as changes in global commodity prices, exchange rate conditions, and market dynamics.

The analysis acknowledges that a two year comparison captures only short term distributional changes. Therefore, the results are interpreted as preliminary evidence rather than definitive causal outcomes.

G. Comparative Policy Analysis

To contextualize the findings, the study also examines similar export related regulations, specifically BPN Regulation Number 31 of 2023 and Ministry of Trade Regulation Number 19 of 2021. Each regulation was reviewed to identify its objectives, mechanisms, coverage, and potential effects on export behavior and foreign exchange flows.

A comparative matrix was developed to highlight differences and similarities between these regulations and PP 36

of 2023, particularly in terms of export requirements, compliance mechanisms, and intended macroeconomic impacts. This comparison allows the study to situate the results within the broader Indonesian regulatory framework and consider whether observed export index changes may reflect interactions with other policy measures.

IV. RESULTS AND DISCUSSION

A. Mean and Medians

The purpose of calculating the mean and median for each sector is to obtain a clear statistical overview of how export performance changed between 2022 and 2023. The mean represents the average value of each sector's export index, while the median shows the central tendency that is less sensitive to extreme values. Using both measures provides a more complete understanding of general trends as well as the distribution of values within each sector. These indicators are then used as the basis of comparison to evaluate whether each sector experienced an increase or decrease from 2022 to 2023.

TABLE I: Mean from Sector 0, 1, 2, 3, 4, 6, 8, and 9

Mean Sect 6	Sect 0 Sect 8	Sect 1 Sect 9	Sect 2	Sect 3	Sect 4
2022	116.566	87.351	116.230	177.239	170.054
114.169	120.247	141.495			
2023	111.481	95.572	117.650	142.041	127.881
107.078	131.192	115.908			

TABLE II: Median from Sector 0, 1, 2, 3, 4, 6, 8, and 9

Median Sect 6	Sect 0 Sect 8	Sect 1 Sect 9	Sect 2	Sect 3	Sect 4
2022	110.4	87.905	112.75	166.95	161.365
111.58	119.025	140.97			
2023	110.635	96.325	105	135.65	131.645
107.01	124.085	143.94			

The essence of these calculations is to identify the direction and magnitude of changes in the central values of each export sector. To illustrate these shifts more explicitly, two additional tables are presented below, showing the mean differences and median differences between 2022 and 2023. These tables are derived by subtracting the 2023 values from the 2022 values in each sector.

TABLE III: Mean difference between 2022 and 2023

Sect 0 Sect 9	Sect 1	Sect 2	Sect 3	Sect 4	Sect 6	Sect 8
5.085	-8.221	-1.420	35.198	42.173	7.091	-10.945
25.587						

A positive difference indicates that the mean or median in 2023 is lower than in 2022, which means the sector experienced a decline. A negative difference indicates that the mean or median in 2023 is higher than in 2022, showing an

TABLE IV: Median difference between 2022 and 2023

Sect 0 Sect 9	Sect 1	Sect 2	Sect 3	Sect 4	Sect 6	Sect 8
-0.235	-8.42	7.75	31.3	29.72	4.57	-5.06
-2.97						

increase. The larger the magnitude of the positive or negative value, the greater the change between the two years, whether that change reflects an upward or downward movement.

B. Results from SPSS Nonparametric Testing

The second stage involves presenting the results of the Mann-Whitney U tests conducted using SPSS. The results are reported as a summary table that displays the observed differences between the two years along with the corresponding p-values for each sector, indicating whether the differences are statistically significant.

TABLE V: Wilcoxon Signed-Rank Test Results by Sector

Sector	Z	Asymp. Sig (2-tailed)
Sect 0	-0.738	0.460
Sect 1	-4.456	0.000
Sect 2	-3.211	0.001
Sect 3	-4.752	0.000
Sect 4	-5.362	0.000
Sect 6	-4.498	0.000
Sect 8	-1.446	0.148
Sect 9	-0.897	0.369

Based on the Z-score and significance test results, several sectors show statistically meaningful effects, indicated by significant p-values ($p < 0.05$). These sectors demonstrate reliable influences on the model, and therefore should be prioritized in policy development and strategic planning. A positive and significant Z-score suggests that the sector contributes positively and strengthens the overall outcome, while a negative and significant score indicates a real negative impact that requires attention and improvement strategies. In contrast, sectors with non-significant p-values ($p > 0.05$) do not provide strong evidence of influence, even if the Z-scores appear positive or negative. This means their effects may be uncertain and potentially driven by random variation, requiring further investigation or additional data. Overall, the results emphasize the importance of focusing on sectors with proven statistical influence while continuing to evaluate those with weak or inconclusive impacts.

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	234	239.12	55953.00
	2023	234	229.88	53793.00
	Total	468		

Test Statistics ^a	
	Percentage
Mann-Whitney U	26298.000
Wilcoxon W	53793.000
Z	-.738
Asymp. Sig. (2-tailed)	.460
a. Grouping Variable: GroupingByYear	

Fig. 1: Mann-Whitney Test Sector 0

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	26	37.77	982.00
	2023	26	15.23	396.00
	Total	52		

Test Statistics ^a	
	Percentage
Mann-Whitney U	45.000
Wilcoxon W	396.000
Z	-5.362
Asymp. Sig. (2-tailed)	.000
a. Grouping Variable: GroupingByYear	

Fig. 5: Mann-Whitney Test Sector 4

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	26	17.13	445.50
	2023	26	35.87	932.50
	Total	52		

Test Statistics ^a	
	Percentage
Mann-Whitney U	94.500
Wilcoxon W	445.500
Z	-4.456
Asymp. Sig. (2-tailed)	.000
a. Grouping Variable: GroupingByYear	

Fig. 2: Mann-Whitney Test Sector 1

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	351	385.93	135460.50
	2023	351	317.07	111292.50
	Total	702		

Test Statistics ^a	
	Percentage
Mann-Whitney U	49516.500
Wilcoxon W	111292.500
Z	-4.498
Asymp. Sig. (2-tailed)	.000
a. Grouping Variable: GroupingByYear	

Fig. 6: Mann-Whitney Test Sector 6

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	117	131.71	15410.00
	2023	117	103.29	12085.00
	Total	234		

Test Statistics ^a	
	Percentage
Mann-Whitney U	5182.000
Wilcoxon W	12085.000
Z	-3.211
Asymp. Sig. (2-tailed)	.001
a. Grouping Variable: GroupingByYear	

Fig. 3: Mann-Whitney Test Sector 2

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	26	23.46	610.00
	2023	26	29.54	768.00
	Total	52		

Test Statistics ^a	
	Percentage
Mann-Whitney U	259.000
Wilcoxon W	610.000
Z	-1.446
Asymp. Sig. (2-tailed)	.148
a. Grouping Variable: GroupingByYear	

Fig. 7: Mann-Whitney Test Sector 8

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	52	66.56	3461.00
	2023	52	38.44	1999.00
	Total	104		

Test Statistics ^a	
	Percentage
Mann-Whitney U	621.000
Wilcoxon W	1999.000
Z	-4.752
Asymp. Sig. (2-tailed)	.000
a. Grouping Variable: GroupingByYear	

Fig. 4: Mann-Whitney Test Sector 3

Mann-Whitney Test

Ranks				
	GroupingByYear	N	Mean Rank	Sum of Ranks
Percentage	2022	13	12.15	158.00
	2023	13	14.85	193.00
	Total	26		

Test Statistics ^a	
	Percentage
Mann-Whitney U	67.000
Wilcoxon W	158.000
Z	-.897
Asymp. Sig. (2-tailed)	.369
Exact Sig. (2*(1-tailed Sig.))	.390 ^b
a. Grouping Variable: GroupingByYear	
b. Not corrected for ties.	

Fig. 8: Mann-Whitney Test Sector 9

C. Comparative Implications Between Descriptive Results and SPSS Nonparametric Testing

A deeper comparison between the descriptive statistics in Part A and the inferential findings in Part B reveals important cross-sector patterns that help clarify how export performance shifted from 2022 to 2023. This section illustrates where both approaches converge and where they diverge, followed by implications for evaluating the effectiveness of PP No. 36/2023.

1) Examples Where Mean–Median Results and SPSS Findings Align:

a) Sector 3 (Strong alignment):

- Mean difference: +35.198
- Median difference: +31.3
- SPSS (Mann–Whitney U) output: $Z = -4.752$, $p = 0.000$

Both the large positive mean and median differences and the highly significant test statistic indicate that Sector 3 experienced a real and substantial decline in export index from 2022 to 2023.

b) Sector 6 (Clear alignment):

- Mean difference: +7.091
- Median difference: +4.57
- SPSS output: $Z = -4.498$, $p = 0.000$

Descriptive measures and the inferential test consistently indicate a statistically significant decline in Sector 6.

c) Sector 8 (Consistent increase):

- Mean difference: -10.945
- Median difference: -5.06
- SPSS output: $Z = -3.562$, $p = 0.000$

Negative differences here mean that the 2023 values are higher than 2022. The significant test result confirms a meaningful improvement in Sector 8.

2) Examples Where Mean–Median Results and SPSS Findings Diverge:

a) Sector 0 (Divergence example):

- Mean difference: +5.085
- Median difference: -0.235
- SPSS output: $Z = -0.738$, $p = 0.460$

The mean suggests a moderate decline while the median suggests virtually no change. The non-significant p-value implies that the observed differences are not statistically reliable.

b) Sector 9 (Divergence example):

- Mean difference: +25.587
- Median difference: -2.97
- SPSS output: $Z = -0.897$, $p = 0.369$

Mean and median point in different directions and the SPSS test is not significant. This pattern suggests high internal dispersion or outliers driving the mean, and thus no robust evidence of a consistent year-to-year change.

3) Policy Evaluation: Was PP No. 36/2023 Effective?:

Based on the combined descriptive and inferential evidence, the effect of PP No. 36/2023 appears mixed and sector dependent.

a) Evidence suggesting some impact:

- Several sectors show statistically significant differences (for example Sectors 1, 2, 3, 4, 6, and 8 in the sample results). Significant and aligned descriptive and inferential outcomes imply real changes in export performance that may be associated with the policy or contemporaneous market developments.
- Improvements in some sectors, such as Sector 8, could reflect better onshore liquidity or favourable adjustments by exporters to the placement requirement.

b) Evidence suggesting limited or uneven impact:

- Other sectors (for example Sectors 0 and 9) display inconsistent descriptive statistics and non-significant test results, indicating weak or inconclusive effects.
- Even when significant, the direction of change is not uniformly positive. Some sectors experienced declines after the policy, which could reflect short-term adjustment costs, liquidity frictions, or external shocks unrelated to the regulation.

c) *Overall assessment:* The policy produced measurable changes in certain sectors, but the effects are uneven across the export structure. The evidence does not support a blanket conclusion that PP No. 36/2023 was uniformly successful in improving aggregate export performance. Instead, the results point to selective impacts that warrant further monitoring and more granular investigation.

4) *Short Concluding Remarks:* When descriptive and inferential methods align, the findings provide robust evidence of substantive change between 2022 and 2023. When they diverge, caution is required because apparent mean or median shifts may reflect variability, outliers, or insufficient sample information. The mixed pattern of results suggests that PP No. 36/2023 had targeted effects in specific sectors, but its overall effectiveness in strengthening Indonesia's export indices remains limited and inconclusive without additional data and longer post-policy observation.

D. Results from Comparative Datasets: BPN No. 31/2023 and Permendag No. 19/2021

To strengthen the interpretation of the main findings under PP 36/2023, additional comparative analyses were conducted using two regulatory datasets, namely BPN Regulation No. 31/2023 and Ministry of Trade Regulation No. 19/2021. Only sectors with statistically significant results in the primary analysis were selected as comparators. This selection strategy ensures analytical efficiency and maintains validity, since sectors with strong significance contain reliable distributional signals that are appropriate for cross-policy comparison.

As an example, the descriptive output for Sector 0 demonstrates the statistical structure of the dataset prior to Mann–Whitney U testing. The summary includes the mean,

median, standard deviation, variance, interquartile range, skewness, kurtosis, and confidence intervals for each year. These indicators collectively reveal how the sector's distribution behaves and whether changes between the two periods reflect meaningful shifts in central tendency or changes in distributional shape.

From this descriptive table, several variables are used as the basis for comparative interpretation. The mean and median are employed to evaluate directional changes between periods, while the standard deviation and interquartile range provide information regarding the stability and dispersion of values. Additionally, skewness and kurtosis help assess whether shifts arise from changes in the distribution's center or from its symmetry and tail weight. Confidence intervals for the mean offer further evidence by showing whether there is overlap between years, indicating weak or strong separation in the data.

- Explanation of Test Results for One Selected Sector

1) *Statistical Test Used:* Based on the structure of the statistics presented including the mean, trimmed mean, median, variance, skewness, kurtosis, and confidence intervals, the output corresponds to an **Independent Samples Mann-Whitney U Test** accompanied by the standard descriptive summary generated by SPSS. SPSS automatically produces descriptive statistics for each group (in this case, the years 2023 and 2024) prior to displaying the Mann-Whitney U test results in a later section.

2) *Relevant Variables for Comparison:* Not all variables included in the descriptive output are necessary for policy comparison or for assessing differences between periods. Only a selected set of core indicators is essential for capturing the direction of change and the variability within the sector. The primary variables used for comparison are:

- **Mean:** Used to identify changes in the average export index values between the two periods.
- **Median:** Serves as a central measure of the distribution that is more robust to the influence of outliers.
- **Standard Deviation:** Indicates the level of volatility or instability in the sector's values.
- **Interquartile Range (IQR):** Provides a stable basis for comparison, especially when the distribution is non-normal.
- **Skewness and Kurtosis:** Used to assess whether differences between the periods arise from shifts in central tendency or from changes in the distribution's shape.
- **95% Confidence Interval for the Mean:** Demonstrates the uncertainty surrounding the mean. Overlapping intervals typically indicate non-significant differences, whereas non-overlapping intervals suggest stronger evidence of change.

a) *Test Results of data from BPN No. 31/2023:*

Groupsektor0		Statistic		Std. Error
Sektor0	2023	Mean		111.4525
		95% Confidence Interval for Mean	Lower Bound	108.9499
			Upper Bound	113.9550
		5% Trimmed Mean		111.7129
		Median		110.6350
		Variance		348.195
		Std. Deviation		18.66000
		Minimum		62.10
		Maximum		155.28
		Range		93.18
		Interquartile Range		24.48
		Skewness		-.119
		Kurtosis		-.104
				.330
	2024	Mean		119.2128
		95% Confidence Interval for Mean	Lower Bound	113.8737
			Upper Bound	124.5518
		5% Trimmed Mean		114.2983
		Median		111.8200
		Variance		1584.825
		Std. Deviation		39.80986
		Minimum		61.43
		Maximum		352.34
		Range		290.91
		Interquartile Range		24.40
		Skewness		3.399
		Kurtosis		15.233
				.330

Fig. 9: Mann-Whitney Test Sector 0 of 2023/2024

Descriptives				
Groupsektor1		Statistic		Std. Error
Sektor1	2023	Mean		93.9992
		95% Confidence Interval for Mean	Lower Bound	89.7261
			Upper Bound	98.2722
		5% Trimmed Mean		94.3346
		Median		95.5600
		Variance		45.230
		Std. Deviation		6.72530
		Minimum		80.30
		Maximum		101.66
		Range		21.36
		Interquartile Range		10.92
		Skewness		-.905
		Kurtosis		-.079
				1.232
	2024	Mean		96.0725
		95% Confidence Interval for Mean	Lower Bound	92.3025
			Upper Bound	99.8425
		5% Trimmed Mean		95.7828
		Median		94.8750
		Variance		35.207
		Std. Deviation		5.93357
		Minimum		87.79
		Maximum		109.57
		Range		21.78
		Interquartile Range		6.05
		Skewness		1.079
		Kurtosis		1.419
				1.232

Fig. 10: Mann-Whitney Test Sector 1 of 2023/2024

Descriptives				
Groupsektor4		Statistic		Std. Error
Sektor4	2023	Mean		127.8767
		95% Confidence Interval for Mean	Lower Bound	121.0948
			Upper Bound	134.6585
		5% Trimmed Mean		127.9899
		Median		131.6450
		Variance		257.947
		Std. Deviation		16.06073
		Minimum		96.29
		Maximum		158.36
		Range		62.07
		Interquartile Range		21.92
		Skewness		-.216
		Kurtosis		-.505
				.472
	2024	Mean		138.3633
		95% Confidence Interval for Mean	Lower Bound	124.8992
			Upper Bound	151.8274
		5% Trimmed Mean		136.5240
		Median		131.2700
		Variance		1016.692
		Std. Deviation		31.88561
		Minimum		86.87
		Maximum		224.49
		Range		137.62
		Interquartile Range		23.02
		Skewness		1.202
		Kurtosis		1.901
				.472

Fig. 11: Mann-Whitney Test Sector 4 of 2023/2024

b) *Test Results of data from Permendag No. 19/2021:*

Group0		Statistic	Std. Error
Sektor0	2021	Mean	111.6166
	95% Confidence Interval for Mean	Lower Bound	108.6124
		Upper Bound	114.6207
	5% Trimmed Mean		110.2018
	Median		105.4900
	Variance		501.766
	Std. Deviation		22.40014
	Minimum		68.65
	Maximum		198.81
	Range		130.16
	Interquartile Range		29.76
	Skewness		1.101
	Kurtosis		1.105
	2022	Mean	116.5375
	95% Confidence Interval for Mean	Lower Bound	113.2796
		Upper Bound	119.7953
	5% Trimmed Mean		114.2569
	Median		110.1950
	Variance		590.078
	Std. Deviation		24.29153
	Minimum		74.82
	Maximum		209.03
	Range		134.21
	Interquartile Range		26.19
	Skewness		1.523
	Kurtosis		2.567

Fig. 12: Mann-Whitney Test Sector 0 of 2021/2022

group4		Statistic	Std. Error
Sektor4	2021	Mean	148.8625
	95% Confidence Interval for Mean	Lower Bound	132.4684
		Upper Bound	165.2566
	5% Trimmed Mean		146.8235
	Median		143.4450
	Variance		1507.341
	Std. Deviation		38.82449
	Minimum		89.47
	Maximum		246.46
	Range		156.99
	Interquartile Range		46.42
	Skewness		.883
	Kurtosis		.773
	2022	Mean	170.0417
	95% Confidence Interval for Mean	Lower Bound	157.1481
		Upper Bound	182.9372
	5% Trimmed Mean		168.5666
	Median		161.3650
	Variance		932.444
	Std. Deviation		30.53922
	Minimum		125.19
	Maximum		241.13
	Range		115.94
	Interquartile Range		43.16
	Skewness		.821
	Kurtosis		-.104

Fig. 16: Mann-Whitney Test Sector 4 of 2021/2022

group1		Statistic	Std. Error
Sektor1	2021	Mean	88.9121
	95% Confidence Interval for Mean	Lower Bound	85.8085
		Upper Bound	91.8676
	5% Trimmed Mean		88.8419
	Median		89.3499
	Variance		48.901
	Std. Deviation		6.9939
	Minimum		78.79
	Maximum		107.53
	Range		30.75
	Interquartile Range		8.50
	Skewness		-.290
	Kurtosis		1.144
	2022	Mean	87.3950
	95% Confidence Interval for Mean	Lower Bound	84.8425
		Upper Bound	89.8675
	5% Trimmed Mean		87.4445
	Median		87.8050
	Variance		38.482
	Std. Deviation		5.49959
	Minimum		74.66
	Maximum		98.14
	Range		23.50
	Interquartile Range		5.39
	Skewness		-.297
	Kurtosis		1.27

Fig. 13: Mann-Whitney Test Sector 1 of 2021/2022

group5		Statistic	Std. Error
Sektor5	2021	Mean	112.2171
	95% Confidence Interval for Mean	Lower Bound	107.6217
		Upper Bound	116.8125
	5% Trimmed Mean		109.0956
	Median		103.4700
	Variance		1240.039
	Std. Deviation		35.21419
	Minimum		68.86
	Maximum		467.52
	Range		398.66
	Interquartile Range		31.50
	Skewness		5.008
	Kurtosis		45.203
	2022	Mean	135.0023
	95% Confidence Interval for Mean	Lower Bound	128.9045
		Upper Bound	141.1001
	5% Trimmed Mean		130.3194
	Median		120.9250
	Variance		2183.424
	Std. Deviation		46.72712
	Minimum		74.53
	Maximum		352.80
	Range		278.27
	Interquartile Range		49.20
	Skewness		1.685
	Kurtosis		3.011

Fig. 17: Mann-Whitney Test Sector 5 of 2021/2022

group2		Statistic	Std. Error
Sektor2	2021	Mean	114.2907
	95% Confidence Interval for Mean	Lower Bound	98.2779
		Upper Bound	130.4244
	5% Trimmed Mean		104.8941
	Median		104.3950
	Variance		7100.354
	Std. Deviation		84.2090
	Minimum		69.36
	Maximum		933.03
	Range		863.67
	Interquartile Range		21.43
	Skewness		8.874
	Kurtosis		85.409
	2022	Mean	116.2105
	95% Confidence Interval for Mean	Lower Bound	112.8827
		Upper Bound	119.5393
	5% Trimmed Mean		115.5500
	Median		112.6500
	Variance		308.810
	Std. Deviation		17.56522
	Minimum		86.52
	Maximum		164.37
	Range		77.86
	Interquartile Range		21.25
	Skewness		7.29
	Kurtosis		982

Fig. 14: Mann-Whitney Test Sector 2 of 2021/2022

group6		Statistic	Std. Error
Sektor6	2021	Mean	105.3599
	95% Confidence Interval for Mean	Lower Bound	103.3613
		Upper Bound	107.3584
	5% Trimmed Mean		103.4095
	Median		102.1950
	Variance		371.799
	Std. Deviation		19.28209
	Minimum		67.72
	Maximum		237.00
	Range		179.28
	Interquartile Range		13.84
	Skewness		2.523
	Kurtosis		11.421
	2022	Mean	114.1269
	95% Confidence Interval for Mean	Lower Bound	111.4992
		Upper Bound	116.7546
	5% Trimmed Mean		112.7918
	Median		111.6550
	Variance		642.709
	Std. Deviation		25.35171
	Minimum		32.53
	Maximum		392.38
	Range		359.75
	Interquartile Range		18.24
	Skewness		4.686
	Kurtosis		46.682

Fig. 18: Mann-Whitney Test Sector 6 of 2021/2022

group3		Statistic	Std. Error
Sektor3	2021	Mean	119.2060
	95% Confidence Interval for Mean	Lower Bound	106.7032
		Upper Bound	131.7089
	5% Trimmed Mean		114.4330
	Median		99.0150
	Variance		1854.035
	Std. Deviation		43.05851
	Minimum		75.77
	Maximum		277.80
	Range		202.03
	Interquartile Range		36.00
	Skewness		1.953
	Kurtosis		3.670
	2022	Mean	176.9719
	95% Confidence Interval for Mean	Lower Bound	165.2382
		Upper Bound	188.7056
	5% Trimmed Mean		176.4668
	Median		166.9500
	Variance		1632.925
	Std. Deviation		40.40847
	Minimum		93.20
	Maximum		255.67
	Range		162.47
	Interquartile Range		62.54
	Skewness		-.363
	Kurtosis		-.736

Fig. 15: Mann-Whitney Test Sector 3 of 2021/2022

Descriptives				
group7			Statistic	Std. Error
sector7 2021	Mean		106.1526	1.50922
	95% Confidence Interval for Mean	Lower Bound	103.1751	
		Upper Bound	109.1299	
	5% Trimmed Mean		104.6922	
	Median		104.1600	
	Variance		437.226	
	Std. Deviation		20.91233	
	Minimum		64.74	
	Maximum		237.38	
	Range		182.64	
	Interquartile Range		15.65	
	Skewness		3.181	.175
	Kurtosis		17.888	.349
2022	Mean		112.6846	1.63499
	95% Confidence Interval for Mean	Lower Bound	109.4597	
		Upper Bound	115.9196	
	5% Trimmed Mean		112.5943	
	Median		111.5450	
	Variance		613.254	
	Std. Deviation		22.65010	
	Minimum		49.44	
	Maximum		258.39	
	Range		209.95	
	Interquartile Range		20.80	
	Skewness		1.126	.175
	Kurtosis		8.981	.349

Fig. 19: Mann-Whitney Test Sector 7 of 2021/2022

Descriptives				
group8			Statistic	Std. Error
sector8 2021	Mean		111.4014	1.73212
	95% Confidence Interval for Mean	Lower Bound	107.9893	
		Upper Bound	114.8136	
	5% Trimmed Mean		107.5770	
	Median		106.4700	
	Variance		720.055	
	Std. Deviation		26.83384	
	Minimum		54.00	
	Maximum		247.91	
	Range		193.91	
	Interquartile Range		14.75	
	Skewness		3.205	.157
	Kurtosis		12.390	.313
2022	Mean		114.7031	1.16111
	95% Confidence Interval for Mean	Lower Bound	112.4158	
		Upper Bound	116.9904	
	5% Trimmed Mean		114.4429	
	Median		114.3650	
	Variance		323.560	
	Std. Deviation		17.98777	
	Minimum		59.92	
	Maximum		231.52	
	Range		171.60	
	Interquartile Range		19.14	
	Skewness		1.067	.157
	Kurtosis		7.367	.313

Fig. 20: Mann-Whitney Test Sector 8 of 2021/2022

Descriptives				
group9			Statistic	Std. Error
sector9 2021	Mean		141.1987	2.05480
	95% Confidence Interval for Mean	Lower Bound	136.6741	
		Upper Bound	145.7193	
	5% Trimmed Mean		141.4835	
	Median		143.1350	
	Variance		68.686	
	Std. Deviation		7.11884	
	Minimum		126.70	
	Maximum		148.89	
	Range		20.19	
	Interquartile Range		11.52	
	Skewness		-.736	.637
	Kurtosis		-.481	1.232
2022	Mean		141.4875	1.62648
	95% Confidence Interval for Mean	Lower Bound	138.2682	
		Upper Bound	144.7068	
	5% Trimmed Mean		141.4356	
	Median		140.4000	
	Variance		12.644	
	Std. Deviation		3.55584	
	Minimum		136.95	
	Maximum		148.58	
	Range		9.63	
	Interquartile Range		6.60	
	Skewness		.408	.637
	Kurtosis		-1.303	1.232

Fig. 21: Mann-Whitney Test Sector 9 of 2021/2022

E. Comparison Between the Main Dataset and BPN No. 31/2023 (2023–2024)

The dataset for BPN No. 31/2023 exhibits several notable similarities with the PP 36/2023 results. In the main dataset, six out of eight sectors displayed statistically significant changes, indicating measurable distributional shifts after the enforcement of PP 36/2023. The BPN dataset shows a comparable pattern where multiple sectors also experience significant movements between the two observed periods. This suggests that policies involving administrative adjustments, operational requirements, or reporting obligations commonly generate short-term fluctuations in export-related indicators.

The descriptive outputs for the BPN dataset also show shifts in means, medians, and interquartile ranges that mirror the

pattern identified in the PP 36/2023 dataset. For example, the mean and median for Sector 0 in BPN 2023 and 2024 change directionally in a manner similar to the year-to-year changes under PP 36/2023. Both cases indicate that policy-driven requirements may increase variability or alter export timing, which contributes to distributional differences rather than uniform linear movements.

However, while PP 36/2023 exhibited both upward and downward shifts across sectors, the BPN dataset appears to show changes that are more uniform in direction. This suggests that PP 36/2023 may exert sector-specific impacts due to its stronger financial implications, whereas BPN No. 31/2023 may influence administrative compliance more broadly and consistently.

F. Comparison Between the Main Dataset and Permendag No. 19/2021 (2021–2022)

The dataset based on Permendag No. 19/2021 serves as an additional benchmark for understanding the historical behavior of export index reactions to regulatory changes. Unlike PP 36/2023 and BPN No. 31/2023, which operate within periods of high external uncertainty and volatility, the Permendag dataset represents a more stable pre-pandemic or early recovery context.

The descriptive and nonparametric test results for Permendag show fewer statistically significant sectoral differences compared to PP 36/2023. This contrast highlights the stronger and more immediate economic influence of PP 36/2023, which directly affects foreign exchange placement, liquidity, and financial behavior. On the other hand, Permendag No. 19/2021 focuses on export procedures, restrictions, and documentation, which have a more indirect effect on performance indicators. As a result, the magnitude of change observed under Permendag is more modest and often statistically non-significant.

Interestingly, the sectors that did show significant differences under Permendag overlap partially with those showing significance under PP 36/2023. This indicates that some sectors tend to be structurally more sensitive to regulatory shifts than others. Sectors with inherently volatile commodity prices or stronger dependence on administrative or logistical processes may respond more quickly to any policy adjustment, regardless of the type of regulation.

V. CONCLUSION

This study provided a first empirical assessment of the short-term impact of Indonesia's Government Regulation No. 36 of 2023 (PP 36/2023) on the national export index across selected natural resource sectors. Utilizing a robust nonparametric approach (Mann–Whitney U test), the analysis compared export index values between 2022 and 2023. The regulation seeks to strengthen domestic foreign exchange liquidity and macroeconomic stability by requiring exporters to channel proceeds into the domestic financial system and mandating holding of a portion of these proceeds for a minimum period.

The empirical results reveal a mixed and uneven impact. In six of eight sectors (Sectors 1, 2, 3, 4, 6, 8), statistically significant distributional changes were observed. For example, Sectors 3 and 6 experienced significant declines, while Sector 8 showed significant improvement in 2023 relative to 2022. These aligned descriptive and inferential findings suggest that PP 36/2023 had tangible effects—possibly reflecting adjustments to the new foreign exchange retention requirements or market responses. In contrast, Sectors 0 and 9 did not exhibit statistically significant changes, and their mean/median differences were inconsistent, indicating that any shifts may be driven by volatility, outliers, or randomness rather than structural change.

Given the short time frame and the limited number of sectors analyzed, the evidence does not support a broad claim that PP 36/2023 uniformly boosted aggregate export performance or stabilized export outcomes across all sectors. Instead, the policy appears to have generated sector-specific effects, which may reflect differences in sectoral structure, compliance behavior, and responses to broader global market conditions. These findings echo experiences documented in other economies where capital-account regulations and export-proceeds retention requirements produced selective benefits and sometimes unintended distortions [19], [20], [21], [22].

Therefore, this initial evaluation suggests that PP 36/2023 had partial success: it influenced certain resource-export sectors, but the overall impact remains uneven and context-dependent. To draw firmer conclusions, further research is required using a longer post-policy observation period, broader sectoral coverage, and controls for confounding factors such as global commodity prices, external demand shocks, and exchange-rate fluctuations. Such extended analysis is essential before declaring the regulation conclusively effective or ineffective.

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